

Abdul Sami Khan

AI Engineer

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EXPERIENCE

BeetleOPS

Nov 2025 – Present

AI Engineer

Jun 2026 – Present

- Driving advanced R&D initiatives, specializing in state-of-the-art deep learning architectures including Transformers and Vision Transformers (ViTs).
- Optimizing and deploying models for edge computing environments, utilizing techniques such as model quantization to reduce memory footprint and latency.
- Evaluating and improving model accuracy and computational efficiency across complex spatial and temporal tasks.
- Collaborating with cross-functional teams to integrate optimized AI models into production APIs and software systems.

AI & Data Science Associate - Surveillance & Edge AI R&D

Nov 2025 – Jun 2026

- Drove end-to-end R&D of intelligent surveillance systems, spanning data creation, AI research, system design, deployment, and analytics.
- Designed and developed scalable computer vision pipelines for detection, tracking, behavior understanding, and multi-camera intelligence in real-time environments.
- Researched, trained, fine-tuned, and benchmarked multiple deep learning architectures for spatial and temporal understanding under edge and latency constraints.
- Built edge-optimized inference systems capable of handling multiple concurrent CCTV/RTSP streams within limited compute and memory budgets.
- Owned the complete data lifecycle: collection strategy, dataset design, annotation workflows, quality control, and iterative dataset improvement.
- Developed AI-integrated applications and services, connecting model inference with system logic, APIs, and downstream consumers.
- Designed and implemented analytics dashboards to visualize surveillance intelligence, activity trends, and system performance for operational decision-making.
- Collaborated with cross-functional teams to transition research prototypes into deployable production systems.

Knowledge Discovery and Data Science (KDD)

Jun 2024 – Present

Deep Learning Research Team Lead

Jun 2024 – Present

- Promoted to lead a team of 4 interns on a PhD-level research initiative for intrusion detection.
- Explored and benchmarked CNN-BiLSTM, Transformers, and Quantum Neural Networks (QNNs).
- Integrated Explainable AI (XAI) techniques (e.g., SHAP) for interpretability.
- Applied Probabilistic Matrix Averaging (PMA) for robust ensemble evaluation.
- **Technologies:** Python, PyTorch, CNNs, Qiskit, BiLSTM, Transformers, QNNs, SHAP.

Data Research Associate

Sep 2024 – Dec 2024

- Worked on Real-Time 2D Tactical Cricket Mapping using YOLOv8, homography, and custom datasets.
- **Technologies:** Python, YOLOv8, OpenCV, Homography, Dataset Curation.

Data Engineer Intern

Jun 2024 – Aug 2024

- Developed PlotPal, a visualization chatbot that generates plots from CSV data via natural language.
- Integrated Flask backend with Gemini API and Matplotlib.
- **Technologies:** Python, Flask, Matplotlib, Gemini API.

EDUCATION

Bachelors in Data Science, FAST NUCES, Pakistan

2022 – Current

A Levels, Westminster School, Pakistan

2019 – 2021

O Levels (6 As, 2 Bs), International College of Arts and Sciences, Pakistan

2017 – 2019

RELEVANT COURSEWORK

Core Machine Learning, Deep Learning, Big Data Analytics, Data Structures, Algorithm Analysis, Advanced Project Management, Advanced Statistics, Game Theory

TECHNICAL SKILLS

- **Languages:** Python, C++, C#, JavaScript, SQL, R, Assembly
- **Frameworks:** PyTorch, Scikit-learn, Flask, Node.js, OpenCV, YOLOv8, Pandas, NumPy, Statsmodels, SHAP, Qiskit
- **Tools:** Git, VS Code, Jupyter, Google Colab, Arduino/PlatformIO, FFmpeg, Roboflow, Label Studio, TradingView
- **Domains:** Edge computing (ESP32, RPi 5), Data pipelines, Algorithmic trading, ML deployment, Computer Vision, NLP, Time-Series Forecasting

SELECTED PERSONAL PROJECTS & RESEARCH

Edge AI Research & Development

2024 – 2025

Personal Research

- Focused on optimizing AI for low-power edge devices.
- On-Device Object Detection: Deployed YOLO-inspired models on ESP32-CAM for low-power inference.
- Tiny LLM on ESP32-S3: Implemented quantized language models for on-device conversational agents.
- Distributed Architecture: Split quantized LLM processing across multiple ESP32s for sequential inference.
- Raspberry Pi 5 Deployment: Deployed DeepSeek R1, optimizing LLMs for edge hardware.

AI-Powered Home Rovers

2024

Robotics Project

- Built autonomous rovers serving as home robot assistants with on-device perception, navigation, and chatbot interaction using ESP32 and ESP32-CAM.

YouTube Automation Pipeline

2024

Content Automation

- Automated workflow to download, analyze, trim, and publish Shorts with AI integration using Gemini AI, yt-dlp, and FFmpeg.

AI for Street Fighter Gameplay

2024

Reinforcement Learning

- Collected pro gameplay data, tested models, and deployed an MLP outperforming Random Forest.

Oracle SQL Chatbot

2024

NLP Project

- Enabled natural-language querying of Oracle SQL*Plus databases via chatbot using Flask and LLMs.

Jarvis - Multi-Agent Virtual Assistant

2023

Automation Project

- Early project: developed a multi-agent chatbot using Python and APIs for task automation and scheduling.

ACADEMIC PROJECTS

- **Inflation Forecasting:** Analyzed WDI data and benchmarked forecasting models (ARIMA, Ridge, LASSO, Elastic Net), comparing via MSE.
- **Demand Forecasting:** Performed cluster analysis and ML forecasting on electricity demand data; built web interface for visualization.
- **Multi-Agent Sim:** Implemented reinforcement-learning agents reaching Nash equilibrium with evolutionary strategy updates (Game Theory).
- **Vendor System:** Built full-stack vendor/contract management system with MySQL, Node.js, and Flask.
- **Data Viz Projects:** Developed multiple dataset visualizations using D3.js, Tableau, and Power BI for coursework.
- **Sorting Visualizer:** Created an interactive visualization of sorting algorithms in C++ to demonstrate DSA concepts.

- **Cafe Management:** Developed an OOP-based end-to-end order and inventory management system in C++.
- **Chess Game:** Developed a complete chess game from scratch in C++, demonstrating strong OOP concepts.
- **Brick Breaker:** Implemented a game in Assembly Language with multiple levels and difficulty settings.
- **Hardware Calc:** Built a calculator using multiplexers and logic components (Digital Logic Design).

ACHIEVEMENTS

- Winner - Nascon Trading Tournament (2023): Secured 1st place, judged by Chief of Pakistan Stock Exchange; prize Rs.40,000.
- Victorious in two gaming tournaments: Rocket League and Tekken 7.
- Awarded 100% scholarship for A Levels after achieving 6 As and 2 Bs in O Levels.